

1. ECOSYSTEMS

Classification, Structure & Function of Ecosystem

Q.116 Four ecosystems have been given, out of which three are alike in some manner and one is different. Select the one that is different with respect to the nature of the ecosystem.

- (A) River (B) Forest
(C) Aquarium (D) Village

Ans : (C)

[PCB Civil 2022]

Food Chains, Food Webs & Ecological Pyramids

Q.119 Which of the following heterotrophs is classified as a secondary consumer?

- (A) Rodents (B) Cows
(C) Goats (D) Sparrows

Ans : (D)

[PCB Civil 2022]

2. NATURAL RESOURCES

Water Resources: Surface & Ground Water Utilization, Floods & Droughts

Q.17 The horizontal tunnels constructed at shallow depths along the banks of a river to intercept ground water tables are called_____.

- (A) canals (B) infiltration galleries
(C) springs (D) lakes

Ans : (B)

[PCB Civil 2022]

Q.56 The line joining points of equal depths of evapotranspiration is _____.

- (A) isohyets (B) isopluvial
(C) isopleth (D) isothermal

Ans : (C)

[PCB Civil 2022]

Q.57 The daily flows of a river have 1050 data points, and the number of days the flow being equal to or greater than its class value is 205. Find the percentage probability of the flow magnitude being equalled or exceeded.

- (A) 0.15 (B) 0.195
(C) 0.255 (D) 0.295

Ans : (B)

[PCB Civil 2022]

Q.58 Which of the following is the Aridity Index (AI)?

PET = Potential evapotranspiration

AET = Actual evapotranspiration

- (A) $\times$ AET
(B) AET
(C) PET
(D) $\frac{V}{PET}$ Hints : CE_Hydrology_BG_AKK_22 Items_Translation Template-TM Aridity index = $\frac{PET-AET}{PET} \times 100$ PET K Subramanya Engineering hydrology 4, 212 McGrawHillEducationPrivateLimited

Ans : None

[PCB Civil 2022]

Q.59 interval
Discharge
15 25 32 18 12
m³/s

(A) 30.33%**(B)** 22.33%**(C)** 29.49%**(D)** V 33.33% Hints : CE_Hydrology_BG_AKK_22 Items_Translation Template-TM N=5 M=2 Pp= N+1 100% 100 = 33.3%
5+1 K Subramanya Engineering hydrology McGraw HillEducationPrivateLimited 4, 165**Ans : None**

[PCB Civil 2022]

Q.60 The precipitation falling directly over the streams is called:**(A)** channel precipitation**(B)** throughfall**(C)** interception**(D)** runoff**Ans : (A)**

[PCB Civil 2022]

Q.61 In the chart of weighing bucket rain gauge, the pen had reversed its path of travel at 4 hours of the next day after recording 100 mm of rainfall and the rainfall recorded in the next day at next two hours is 50 mm, then the cumulative rainfall recorded by gauge at say 6 hours of the next day is:**(A)** 150 mm**(B)** 50 mm**(C)** 200 mm**(D)** 100 mm**Ans : (A)**

[PCB Civil 2022]

Q.62 Rate of transpiration depends upon the:**(A)** base period of the plant**(B)** harvesting period of the plant**(C)** Atmospheric humidity and wind**(D)** sowing period of the plant**Ans : (C)**

[PCB Civil 2022]

Q.63 Match the characteristics of catchment.

1. Stream density A. Elevation difference between the catchment outlet and the highest point on the basin perimeter
 2. Drainage density B. Ratio of number of streams of all orders to the area of basin
 3. Relief C. Ratio of total length of streams of all orders within the basin to the area of basin
- Reflecting the pattern of branches that unite to form the trunk stream

(A) 1 - A, 2 - B, 3 - C**(B)** 1 - A, 2 - D, 3 - C**(C)** 1 - B, 2 - D, 3 - A**(D)** 1 - B, 2 - C, 3 - A**Ans : (D)**

[PCB Civil 2022]

Q.64 Two channels of order 2 and order 3 joined together produces an order of which of the following streams?**(A)** 3rd order**(B)** 2nd order**(C)** 1st order**(D)** 5th order**Ans : (A)**

[PCB Civil 2022]

Q.65 What is the average value of duty for crop rice in India?**(A)** 775 hectares/cumec**(B)** 1000 hectares/cumec**(C)** 1500 hectares/cumec**(D)** 1275 hectares/cumec**Ans : (A)**

[PCB Civil 2022]

Q.66 A canal commands an irrigated area of 500 hectares. The intensity of irrigation of paddy in this area is 50%. The transplantation of paddy requires 30 days and total depth of water required by the paddy is 50cm on the field during transplantation. The available water during transplantation on the field is 15 cm. Calculate the duty of water, on the field for the given details.

(A) 730.5 hectares/cumec

(B) 750 hectares/cumec

(C) 740.5 hectares/cumec

(D) 745.5 hectares/cumec

Ans : (C)

[PCB Civil 2022]

Q.135 In India, the south western monsoon called 'kharif' spans mostly from _____.

(A) July to October

(B) April to June

(C) October to March

(D) January to March

Ans : (A)

[PCB Civil 2022]

Q.136 Which of the following crops is classified as Rabi crop in India?

(A) Linseed

(B) Sorghum

(C) Rice

(D) Maize

Ans : (A)

[PCB Civil 2022]

Q.137 Calculate the delta for a crop if its duty is 1000 hectares/cumec on the field. Consider the base period of crop as 100 days.

(A) 74.8 cm

(B) 86.4 cm

(C) 100 cm

(D) 106.56 cm

Ans : (B)

[PCB Civil 2022]

Q.138 Identify whether the given statements related to diversion head works constructed at the head of the canal are correct or incorrect. Statements: A) If the major part or entire ponding of water is achieved by a raised crest and a smaller part or nil part of it is achieved by the shutters, then this barrier is called a barrage. B) If the major part or entire ponding of water is achieved by a raised crest and a smaller part or nil part of it is achieved by the shutters, then this barrier is called a weir.

(A) Statement A is correct but B is incorrect

(B) Statement B is correct but A is incorrect

(C) Both statements are correct

(D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

Q.139 Identify whether the given statements related to diversion head works are correct or incorrect. Statements: A) Fish ladder arrangement is provided in the weir or dam for interrupting/preventing the movement of fishes either to the upstream or downstream. B) Silt ejector is used for removal of silt which has entered the canal.

(A) Statement A is correct but B is incorrect

(B) Statement B is correct but A is incorrect

(C) Both statements are correct

(D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

5. FLUID MECHANICS AND HYDRAULICS

Fluid Dynamics: Continuity Equation, Bernoulli's Equation & Impulse Momentum Equation

Q.28 Which of the following formulas is used to find Chezy's constant?

(A) Ryve's equation

(B) Hazen's equation

(C) Bazin's equation

(D) Continuity equation

Ans : (C)

[PCB Civil 2022]

Q.129 Which of the following is the correct expression, in case of open channel flow, to measure energy at a particular point in the channel?

- (A) 62 where, $E = \text{specific energy}$, $y = \text{Pressure head}$ velocity head
 (B) $E = 2y$ where, $E = \text{specific energy}$, $y = \text{Pressure head}$ 29g velocity head.
 (C) $E = y^2 / 2g$ where, $E = \text{specific energy}$, $y = \text{Pressure head}$ 2g velocity head.
 (D) $E =$ where, $E = \text{specific energy}$, $y = \text{Pressure head}$ velocity head

Ans : None

[PCB Civil 2022]

Q.130 Calculate the specific energy at a point in a rectangular open channel whose width is 2 m, depth 1 m and discharge 2 m³/sec. Take acceleration due to gravity as 10 m/sec².

- (A) 1.05 m (B) 1.20 m
 (C) 1.50 m (D) 1.95 m

Ans : (A)

[PCB Civil 2022]

Q.131 A lined channel of trapezoidal section carries a discharge of 6 m³/sec, at a depth of 1 m, with bottom width 2 m and side slope of 1.5 horizontal to 1 vertical. Consider uniform flow and calculate the velocity of flow.

- (A) 1.312 m/sec (B) 1.714 m/sec
 (C) 2.135 m/sec (D) 2.426 m/sec

Ans : (B)

[PCB Civil 2022]

Q.132 Identify whether the given statements related to basic assumptions in gradually varied flow analysis are correct or incorrect. Statements: A) The head loss in a reach may be computed using an equation applicable to uniform flow, having the same velocity and hydraulic mean radius of the section. B) Channel bottom slope is large.

- (A) Statement A is correct but B is incorrect
 (B) Statement B is correct but A is incorrect
 (C) Both statements are correct
 (D) Both statements are incorrect

Ans : (A)

[PCB Civil 2022]

Q.133 Which of the following is an example of steady rapidly varied flow with broken flow profile?

- (A) The backwaters produced by a dam
 (B) Hydraulic jump
 (C) The flow of water in a pipe of constant diameter at constant velocity
 (D) The flow of water through pipe of constant diameter connected to a pump pumping at a constant rate which is then switched off

Ans : (B)

[PCB Civil 2022]

Q.134 Identify whether the given statements related to the applications of a hydraulic jump are correct or incorrect. Statements: A) It is used to lower the water level on the d/s side of a metering flume and thus maintain low water level in the channel for irrigation, etc. B) It is used to dissipate energy in the water flowing over the spillway and other hydraulic structures and thus prevent scouring downstream.

- (A) Statement A is correct but B is incorrect
 (B) Statement B is correct but A is incorrect
 (C) Both statements are correct
 (D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

Elementary Boundary Layer Theory

Q.126 Who among the following introduced the concept of boundary layer and derived the equations for the boundary layer flow by correct reduction of Navier-Stokes equations?

- (A) Ludwig Prandtl (B) Dickens
 (C) Ryve (D) Von Mises

Ans : (A)

[PCB Civil 2022]

- Q.127** Identify whether the given statements related to application of the boundary layer theory are correct or incorrect. Statements: A) The boundary-layer theory is valid beyond the point of separation. B) Mathematically, the application of the boundary-layer theory converts the character of governing Navier-Stroke equations from elliptic to parabolic.
- (A) Statement A is correct but B is incorrect
 (B) Statement B is correct but A is incorrect
 (C) Both statements are correct
 (D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

- Q.128** Identify whether the given statements related to point of separation and the boundary layer theory in fluids are correct or incorrect. Statements: A) The boundary layer theory does not speak about the location of the point of separation. B) At the point of separation, the boundary layer thickness becomes quite large for the thin layer approximation to be valid.
- (A) Statement A is correct but B is incorrect
 (B) Statement B is correct but A is incorrect
 (C) Both statements are correct
 (D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

7. AIR, WATER, SOIL AND NOISE POLLUTION AND CONTROL TECHNOLOGIES

Air Pollution: Primary & Secondary Pollutants, Ambient Air Quality Standards

- Q.35** Collection efficiency of an electrostatic precipitator ranges between _____ for particles size between $0.1\mu\text{m}$ to $0.2\mu\text{m}$.
- (A) 99.8% to 100% (B) 96% to 97%
 (C) 86% to 88% (D) 83% to 85%

Ans : (B)

[PCB Civil 2022]

- Q.36** The collection efficiency of an electrostatic precipitator under ideal conditions is generally estimated by using_____
- (A) Vander Waals equation (B) Dobson equation
 (C) Deutsch-Anderson equation (D) Schmutzdecke equation

Ans : (C)

[PCB Civil 2022]

- Q.37** A venture scrubber used to control air pollution is an example of_____.
- (A) dry scrubbers (B) wet scrubbers
 (C) gravity scrubbers (D) pressure scrubbers

Ans : (B)

[PCB Civil 2022]

Water Pollution: Sources, Types & Drinking Water Quality Standards

- Q.12** Which of the following property of water is determined by EDTA (ethylene-diaminetetraacetate) titration?
- (A) hardness in water (B) turbidity in water
 (C) dissolved oxygen in water (D) residual chlorine in water

Ans : (A)

[PCB Civil 2022]

- Q.18** Copper sulfate is most commonly used(added) to control_____in lakes.
- (A) chlorine demand of water (B) growth of algae
 (C) dispersion of silt (D) suspended particles

Ans : (B)

[PCB Civil 2022]

- Q.19** As per IS 10500:2012, the mximum permissible limit for fluoride content present in drinking water shall not exceed _____in the absence of alternate source.

- (A) 0.015 ppm
(C) 15 ppm

- (B) 1.5 ppm
(D) 150 ppm

Ans : (B)

[PCB Civil 2022]

Q.20 B-coli or E-coli are harmless organisms, but their presence in water indicates the:

- (A) presence of pathogenic bacteria
(B) absence of pathogenic bacteria
(C) presence of non-pathogenic bacteria
(D) absence of non-pathogenic bacteria

Ans : (A)

[PCB Civil 2022]

Q.34 When carbon monoxide reacts with hemoglobin, it forms_____.

- (A) carboxy-haemoglobin
(B) carbomino-haemoglobin
(C) amino acid
(D) carbon di oxide and hydrogen

Ans : (A)

[PCB Civil 2022]

Q.117 Select the INCORRECT statement with respect to permanent hardness of water.

- (A) It can be eliminated by boiling.
(B) It forms insoluble scum with soap.
(C) It affects the cleaning action of soap.
(D) It contains dissolved salts of calcium and magnesium.

Ans : (A)

[PCB Civil 2022]

Wastewater Treatment: Primary, Secondary & Tertiary Treatment

Q.13 The cleaning of slow sand filter is done by:

- (A) scrapping
(C) air cleaning
- (B) back washing
(D) mechanical staining

Ans : (A)

[PCB Civil 2022]

Q.14 Which of the following comparison is correct with respect to rate of filtration in water purification?. Where SSF=slow sand filter, RSF=rapid sand filter, DMF=dual media filter and MMF=mixed media filter.

- (A) $MMF \geq RSF > DMF > SSF$
(C) $DMF < MMF \geq RSF < SSF$
- (B) $MMF \geq DMF > RSF > SSF$
(D) $DMF \geq MMF < RSF > SSF$

Ans : (B)

[PCB Civil 2022]

Q.15 In a water treatment process by sedimentation the flocculent particles follows_____.

- (A) Type - I settling
(C) Type - III settling
- (B) Type - II settling
(D) Type - IV settling

Ans : (B)

[PCB Civil 2022]

Q.16 Which of the following statements about the tube settler is INCORRECT?

- (A) It is the combination of multiple channels at an angle of 60°.
(B) Settling depth is less than conventional one.
(C) Settling time is less than conventional one.
(D) Effective settling area is decreased by providing multiple tubular channels sloped at an angle of 60° and adjacent to each other.

Ans : (D)

[PCB Civil 2022]

Q.26 In a water distribution system using circular system, the main pipe line will be_____.

- (A) in the centre of the city
- (B) at any one side of city
- (C) on the periphery of the city
- (D) anywhere in the city

Ans : (C)

[PCB Civil 2022]

Q.27 The rugosity coefficient value used in the design of sewer lines depends on_____.

- (A) longitudinal gradient
- (B) hydraulic radius of pipe
- (C) material used in sewer pipe
- (D) cross sectional area of sewer pipe

Ans : (C)

[PCB Civil 2022]

Q.29 According to general standards for discharge of environmental pollutants part -A : effluents, the maximum chemical oxygen demand for effluents from inland surface water shall be_____

- (A) 250 mg/l
- (B) 300 mg/l
- (C) 350 mg/l
- (D) 500 mg/l

Ans : (A)

[PCB Civil 2022]

Q.102 Read the given statements related to secondary treatment of wastewater and identify whether they are correct or incorrect. Statements: A) Secondary treatment of wastewater can be achieved by chemical unit processes such as chemical oxidation, coagulation-flocculation and sedimentation and chemical precipitation. B) Secondary treatment of wastewater cannot be achieved by employing biological processes (aerobic) where bacteria is used as a catalyst for removal of pollutant.

- (A) Statement A is correct but B is incorrect
- (B) Statement B is correct but A is incorrect
- (C) Both statements are correct
- (D) Both statements are incorrect

Ans : (C)

[PCB Civil 2022]

Q.103 The term 'Sludge Volume Index' (SVI), used for the determination of sludge settleability is defined as:

- (A) volume occupied in ml by one gram of solids in the mixed liquor after settling for 30 min
- (B) volume occupied in ml by one gram of solids in the mixed liquor after settling for 150 min
- (C) percentage decrease in the volume of sludge after secondary treatment of waste water
- (D) mass of sludge occupied in 1 ml of mixed liquor after settling for 150 min

Ans : (A)

[PCB Civil 2022]

Q.104 Read the given statements related to biological treatment of waste water and identify whether they are correct or incorrect. Statements: A) The objective of the biological treatment of wastewater is NOT to remove nutrients such as nitrogen and phosphorous from the wastewater. B) The objective of the biological treatment of wastewater is to remove organic matter from the wastewater, which is present in soluble and colloidal form.

- (A) Statement A is correct but B is incorrect
- (B) Statement B is correct but A is incorrect
- (C) Both statements are correct
- (D) Both statements are incorrect

Ans : (B)

[PCB Civil 2022]

Noise Pollution: Sources, Health Hazards & Standards

Q.105 Identify whether the given statements related to Noise Exposure Forecast (NEF) are correct or incorrect. Statements: A) NEF develops a noise contour diagram. B) NEF is associated with road traffic noise.

- (A) Statement A is correct but B is incorrect
- (B) Statement B is correct but A is incorrect
- (C) Both statements are correct
- (D) Both statements are incorrect

Ans : (A)

[PCB Civil 2022]

Q.106 Identify whether the given statements related to noise barrier are correct or incorrect. Statements: A) series of trees between the source and receiver of a noise will decrease the sound level of the noise at the receiver point. B) For high-frequency sound, when diffraction occurs due to a noise barrier, the diffraction angle will be very large.

- (A) Statement A is correct but B is incorrect
- (B) Statement B is correct but A is incorrect
- (C) Both statements are correct
- (D) Both statements are incorrect

Ans : (A)

[PCB Civil 2022]

Q.107 Which of the following 'combinations of noise exposure' has the maximum noise dose?

- (A) An average noise level of 90 dBA during a 4-hour period
- (B) An average noise level of 90 dBA during an 8-hour period
- (C) An average noise level of 80 dBA during a 6-hour period
- (D) An average noise level of 85 dBA during an 8-hour period

Ans : (B)

[PCB Civil 2022]

Q.108 Which of the following is the auditory effect caused by noise pollution?

- (A) Permanent hearing loss
- (B) Loss in working efficiency
- (C) Reduced depth and quality of sleep
- (D) Interference with speech communication

Ans : (A)

[PCB Civil 2022]

Q.109 Which of the following is the non-auditory effect caused by noise pollution?

- (A) Permanent hearing loss
- (B) Loss in working efficiency
- (C) Tinnitus
- (D) Whistling and buzzing in ears

Ans : (B)

[PCB Civil 2022]

Q.110 Identify whether the given statements related to Noise-Induced Hearing Loss (NIHL) are correct or incorrect. Statements: A) NIHL can be caused by a one-time exposure to an intense 'impulse' sound, such as an explosion. B) NIHL can be caused by continuous exposure to loud sounds over an extended period of time, such as noise generated in a woodworking shop.

- (A) Statement A is correct but B is incorrect
- (B) Statement B is correct but A is incorrect
- (C) Both statements are correct
- (D) Both statements are incorrect

Ans : (C)

[PCB Civil 2022]

8. SOLID AND INDUSTRIAL WASTE MANAGEMENT

Municipal Solid Waste Management: Collection, Segregation & Transport

Q.38 Which of the following wastes is NOT generated from domestic processes?

- (A) Coconut shell
(C) Garden trimming

- (B) Food waste
(D) scrap lumber

Ans : (D)

[PCB Civil 2022]

Solid Waste Processing & Disposal Technologies

Q.30 Sludge thickening will reduce the volume of sludge by_____ conditioning

- (A) physical
(C) biological

- (B) chemical
(D) physico-chemical

Ans : (A)

[PCB Civil 2022]

Q.31 Fresh sludge has moisture content of 99%, and after thickening, its moisture content is reduced to 96%.The reduction in the volume of sludge is:

- (A) 197%
(C) 75%

- (B) 5%
(D) 3%

Ans : (C)

[PCB Civil 2022]

Q.32 The two main gases liberated from an anaerobic sludge digestion tank would include:

- (A) NH₃ and CH₄
(C) CH₄ and Ne

- (B) CO₂ and CH₄
(D) NH₃ and CO₂

Ans : (B)

[PCB Civil 2022]

Q.33 On what media does the sludge drying bed work?

- (A) Sand
(C) Geotextile cloth

- (B) polypropylene membrane
(D) Plastic

Ans : (A)

[PCB Civil 2022]

Q.111 Which of the following is the major gas produced from landfill and can be used as a renewable energy?

- (A) Carbon monoxide
(C) Oxygen

- (B) Methane
(D) Hydrogen disulphide

Ans : (B)

[PCB Civil 2022]

Q.112 As per CPHEEO-2016, the minimum siting distance of land for sanitary landfill sites shall be _____ away from the centre line of a highway or railway line and water supply wells.

- (A) 100 m
(C) 1000 m

- (B) 500 m
(D) 1500 m

Ans : (B)

[PCB Civil 2022]

Q.113 Which of the following is NOT an advantage of Extended Producer Responsibility (EPR) policy?

- (A) Creation of incentives for environmentally friendly product designs
(B) Provision of a monetary incentive to the consumer to return the product or package
(C) Increase in natural resource demands for packaging and product containers
(D) Creation of infrastructure for collection and recycling of material

Ans : (C)

[PCB Civil 2022]

9. GLOBAL ENVIRONMENTAL PROBLEMS, POLICIES AND LEGISLATIONS

Indian Environmental Protection Act (EPA 1986)

Q.101 As per Environment (Protection) Rules, 1986, following the general standards for discharge of environmental pollutants part-A : effluents, in the case of disposal to inland surface water, the temperature of disposing effluent

water shall NOT exceed _____ above the receiving water temperature.

- (A) 5°C (B) 10°C
(C) 15°C (D) 20°C

Ans : (A)

[PCB Civil 2022]

Q.114 The abbreviation 'MoEF&CC' is used to indicate a ministry dealing with Environmental Impact Assessment (EIA) studies in India. What does the abbreviation stand for?

- (A) Ministry of Environment, Forest and Climate Change
(B) Ministry of Ecology, Forests and Climate Correction
(C) Ministry of Environment, Foundry, Charcoal and Carbon
(D) Ministry of Ecology, Foundry and Civil Construction

Ans : (A)

[PCB Civil 2022]

10. ENVIRONMENTAL IMPACT ASSESSMENT TOWARDS SUSTAINABLE FUTURE

EIA Structure & Baseline Data Acquisition Methods

Q.115 Identify the INCORRECT statement with respect to public participation in Environmental Impact Assessment (EIA) studies.

- (A) Public participation in EIA reduces communication gap between the experts and locals.
(B) Public participation will be able to provide advice on management, considering local conditions.
(C) Opposition to management will be reduced if people are informed.
(D) Public participation in EIA should be avoided as much as possible to reduce nuisance creation.

Ans : (D)

[PCB Civil 2022]

Risk Assessment Strategies & Environmental Management Plan (EMP)

Q.118 Which of the following audits is a type of management audit that comes under environmental audit categories?

- (A) Compliance audit (B) Policy audit
(C) Acquisition audit (D) Health and safety audit

Ans : (B)

[PCB Civil 2022]

Sustainable Development, Population Explosion & Consumerism

Q.120 Which of the following is an approach to design for sustainability?

- (A) Selection of resources with low environmental impact
(B) Design for safety and reliability
(C) Design for manufacturability
(D) Design for supply chain

Ans : (A)

[PCB Civil 2022]

Q.122 According to E Barbier, the primary objective of sustainable development is:

- (A) to reduce the absolute poverty of the world's poor by providing lasting and secure livelihoods that minimise resource depletion, environmental degradation, cultural disruption and social instability
(B) to meet the needs of the present without compromising the ability of future generations to meet their own needs
(C) equitable use of resources for meeting the needs of the present and future generations without causing damage to the environment
(D) to conserve and nurture the biological diversity, gene pool and other resources for long-term food security

Ans : (A)

[PCB Civil 2022]

Q.123 Select the INCORRECT statement with respect to carbon offset, which is used as a financial instrument in sustainable development.

- (A) Carbon offset aims at reduction in greenhouse gas emissions.
 (B) Use of renewable sources of energy is one of the ways to generate carbon offset.
 (C) One carbon offset is equal to reduction of 1 metric ton of CO₂ or its equivalent in other greenhouse gases.
 (D) One metric ton of CO₂ reduction is equivalent to a reduction of around 450 kg carbon.

Ans : (D)

[PCB Civil 2022]

Q.125 Which of the following is NOT a sustainable development goal targeted to be achieved by 2030, according to the General Assembly of the United Nations?

- (A) No poverty
 (B) Clean water and sanitation
 (C) Quality education
 (D) Life on Mars

Ans : (D)

[PCB Civil 2022]

Life Cycle Assessment (LCA)

Q.121 Which of the following ISO provisions describes the principles and framework for life cycle assessment (LCA)?

- (A) ISO 14040:2006
 (B) ISO 23362:2006
 (C) ISO 15649:2006
 (D) ISO 17985:2006

Ans : (A)

[PCB Civil 2022]

Q.124 Which of the following organisations was the first to work on life cycle assessment?

- (A) The Coca-Cola Company
 (B) PepsiCo
 (C) 7UP Company
 (D) The Peelle Company

Ans : (A)

[PCB Civil 2022]

11. CIVIL ENGINEERING

Q.1 As per IS 456:2000, the static modulus of elasticity of M25 grade concrete is_____.

- (A) 20000 N/mm²
 (B) 25000 N/mm²
 (C) 200 N/sq mm
 (D) 250 N/sq mm

Ans : (B)

[PCB Civil 2022]

Q.2 As per IS 456:2000, the partial factor of safety to be considered for steel reinforcement to assess the strength of a steel reinforced cement concrete structure, according to limit state method of design is_____.

- (A) 1.5
 (B) 1.15
 (C) 1.65
 (D) 2

Ans : (B)

[PCB Civil 2022]

Q.3 Which of the following statements are INCORRECT? Statement 1: Microsilica is a chemical admixture used in making concrete. Statement 2: Microsilica particles are spherical in shape. Statement 3: Mean particle size of micro silica ranges between 10 to 90 microns. Statement 4: Specific surface area of micro silica is in the range of 20000 m²/kg.

- (A) Statements 1 and 2 are incorrect
 (B) Statements 1 and 3 are incorrect
 (C) Statements 2 and 3 are incorrect
 (D) Statements 3 and 4 are incorrect

Ans : (B)

[PCB Civil 2022]

Q.4 As per IS 10262 : 2019, if angular coarse aggregates are replaced by sub-angular coarse aggregates in normal concrete, the quantity of mixing water per unit volume of concrete can be _____ to produce the same workability as that of angular coarse aggregates.

- (A) reduced by 10 kg
(C) increased by 10 kg

- (B) reduced by 20 kg
(D) increased by 20 kg

Ans : (A)

[PCB Civil 2022]

Q.5 As per National building code of India: 2016 (volume 1- part 4), Which of the following civil engineering facility is classified as Group H buildings?

- (A) Warehouses
(C) Oil refineries

- (B) Railways
(D) Highways

Ans : (A)

[PCB Civil 2022]

Q.6 The critical path for the given network diagram is

- (A) A-B-C-H
(B) V-A-D-F-H
(C) A-D-F-G-I

(D) A-D-E-I
Hints : Civil Engg_K M Chaitanya_Const Material Mgmt_Set 1&2 Critical Path is the path in a network connecting initial and final nodes and has a maximum duration. Path A -B-C-H has a duration of 18 days. Path A-D-F-G-I has a duration of 18 days. Path A - D - E - I has a duration of 15 days. Path A - D - F - H has a duration of 19 days. Hence the solution.

Ans : None

[PCB Civil 2022]

Q.7 As per IS 456-2000, what is the value of slump for degree of workability 'High'? Consider the placing condition as Tremie concrete.

- (A) 25 – 50 mm
(C) 10 – 150 mm

- (B) 50 – 100 mm
(D) 100 – 150 mm

Ans : (D)

[PCB Civil 2022]

Q.8 D-cracking on concrete slabs is caused due to _____.

- (A) thermal expansion
(B) surface wear
(C) freezing and thawing
(D) crystallisation of salts in pores

Ans : (C)

[PCB Civil 2022]

Q.9 Pre-planning phase of a construction project does NOT involve _____.

- (A) formulation of general framework
(B) cost-benefit analysis
(C) investment alternatives
(D) placing and tying reinforcement

Ans : (D)

[PCB Civil 2022]

Q.10 The function of management that involves proper communication, leadership and motivation to accomplish the set objectives is _____.

- (A) coordination
(C) directing

- (B) controlling
(D) organising

Ans : (C)

[PCB Civil 2022]

Q.11

- (A) 1 and 0, respectively
(C) 1 and 1, respectively

- (B) 0 and 1, respectively
(D) 0 and 0, respectively

Ans : (D)

[PCB Civil 2022]

Q.21 When centrifugal pump is used to lift water, the direction of flowing water is changed by _____.

(A) 45°

(B) 60°

(C) 90°

(D) 180°

Ans : (C)

[PCB Civil 2022]

Q.22 Which of the following can measure the head loss?

(A) Weir

(B) Notch

(C) Scrubber

(D) Piezometer

Ans : (D)

[PCB Civil 2022]

Q.23 A surge tank is a storage reservoir at the _____ end of a close aqueduct to absorb a sudden _____ of pressure.

(A) downstream, rise

(B) upstream, rise

(C) upstream, fall

(D) downstream, fall

Ans : (A)

[PCB Civil 2022]

Q.24 The valves which are used to remove the sediment in the pipelines are called:

(A) butterfly valves

(B) scour valves

(C) air valves

(D) check valves

Ans : (B)

[PCB Civil 2022]

Q.25 To control the water hammer, which of the following valves will be used?

(A) Air valves

(B) Pilot valves

(C) Scour valves

(D) Blow off valves

Ans : (A)

[PCB Civil 2022]

Q.39 When a retaining wall moves towards the backfill, the pressure exerted on the wall is termed as _____.

(A) passive earth pressure

(B) swelling pressure

(C) pore pressure

(D) active earth pressure

Ans : (A)

[PCB Civil 2022]

Q.40 is wall friction angle, and is inclination of back of wall to the vertical.(A) $= 90^\circ + 0^\circ +$ (B) $90^\circ + \phi + \phi_0 =$ (C) $W = 90^\circ +$ (D) $= 90^\circ -$ - Hints : For Review Civil-SSSJ-Foundation-set1&216ItemsTranslationTemplate Dr.B CPunmia, Soil Mechanics and Foundations, Laxmi Publications Pvt Ltd, 16,562**Ans : None**

[PCB Civil 2022]

Q.41 The load carrying capacity of an individual friction pile is 200 kN. What is the total load carrying capacity of a group of nine such piles with a group efficiency factor of 0.8?

(A) 1800 kN

(B) 1640 kN

(C) 1440 kN

(D) 900 kN

Ans : (C)

[PCB Civil 2022]

Q.42 In the static cone penetration method, for calculation of settlement in cohesionless soils, _____.

- (A) the sand layer is divided into small layers such that each small layer has approximately constant value of the cone resistance
- (B) the sand layer is divided into small layers such that each small layer has different value of the cone resistance
- (C) the sand layer considered to be a single layer so that there is zero value of the cone resistance
- (D) the sand layer is divided into small layers of equal size such that each small layer undergoes same amount of settlement

Ans : (A)

[PCB Civil 2022]

Q.43 The upward pressure developed by the soil on the bottom face of a footing is termed as _____.

- (A) surface tension (B) confining pressure
- (C) contact pressure (D) deviatoric pressure

Ans : (C)

[PCB Civil 2022]

Q.44 Contact pressure Vescription ottootings
distribution diagrams

1. Rigid footing on cohesive soil
2. Flexible footing on cohesive soil
- B.
2. Rigid footing on cohesion-less soil at ground level
4. Flexible footing on cohesionless soil at ground level

- (A) A-3, B-1, C-4, D-2
- (C) A-3, B-2, C-4, D-1

- (B) A-4, B-3, C-2, D-1
- (D) A-4, B-1, C-3, D-2

Ans : (C)

[PCB Civil 2022]

Q.45 In cohesive-frictional soils, to obtain safe load by Static Pile load formula, a factor of safety of _____ in compression and _____ in uplift are recommended respectively.

- (A) 1.5 ; 3 (B) 3; 4.5
- (C) 2.5; 3 (D) 3; 6

Ans : (C)

[PCB Civil 2022]

Q.46 railways, reservoirs, sewerage,
water supply schemes, etc.
D. City Survey 4. To locate the boundaries of
fields, houses, etc.

- (A) A-1; B-3; C-4; D-2
- (C) A-4; B-3; C-2; D-1

- (B) A-1; B-2; C-3; D-4
- (D) A-1; B-4; C-3; D-2

Ans : (A)

[PCB Civil 2022]

Q.47 Which of the following is NOT an advantage of chain surveying?

- (A) It does not require costly equipment.
- (B) It is suitable for small areas with a fair degree of accuracy.
- (C) Computations and plotting are simple.
- (D) It is suitable for large and hilly areas.

Ans : (D)

[PCB Civil 2022]

Q.48 Which of the following is an advantage of chain surveying?

- (A) It is suitable for large areas because chaining for large areas is simple.
 (B) It is suitable for undulated areas.
 (C) It is suitable for small areas with a fair degree of accuracy.
 (D) Results are always accurate.

Ans : (C)

[PCB Civil 2022]

Q.49 The principle of compass surveying is_____.

- (A) direct and indirect ranging
 (B) triangulation
 (C) traversing
 (D) chaining

Ans : (C)

[PCB Civil 2022]

Q.50 If the bearing of line PQ is $N65^{\circ} 35' 00''E$ and the deflection angle (clockwise) between the lines PQ and QR is $34^{\circ} 45' 00''$, then the bearing of line QR is:

- (A) $N30^{\circ} 50'E$ (B) $N34^{\circ} 45'E$
 (C) $S79^{\circ} 40'E$ (D) $N10^{\circ} 20'E$

Ans : (C)

[PCB Civil 2022]

Q.51 Identify the INCORRECT statement about magnetic dip.

- (A) The horizontal angle made by the magnetic needle with vertical is known as dip of the needle.
 (B) Angle of dip varies from 0° at the equator to 90° at the magnetic poles.
 (C) The value of dip varies from place to place.
 (D) Dip angle increases as we go from the equator to the poles.

Ans : (A)

[PCB Civil 2022]

Q.52 The variation in which magnetic meridian swings like a pendulum is known as _____.

- (A) irregular variation (B) annual variation
 (C) secular variation (D) diurnal variation

Ans : (C)

[PCB Civil 2022]

Q.53 Select the option that is correct regarding the following two statements. Statement 1: Local attraction at any station is detected by observing the fore bearing and the back bearing of the line. Statement 2: If the difference between FB and BB is 180° , both the end stations are considered to be free from local attraction.

- (A) Both Statement 1 and Statement 2 are individually true and Statement 2 is the correct explanation of Statement 1.
 (B) Both Statement 1 and Statement 2 are individually true, but Statement 2 is not the correct explanation of Statement 1.
 (C) Statement 1 is true, but Statement 2 is false.
 (D) Statement 1 is false, but Statement 2 is true.

Ans : (A)

[PCB Civil 2022]

Q.54 Given are the bearings of the lines of an open traverse. Correct the bearings by finding local attraction at the affected stations.

PQ: $36^{\circ}00'$ QP: $216^{\circ} 45'$
 QR: $98^{\circ} 15'$ RQ: $276^{\circ}00'$
 RS: $201^{\circ}45'$ SR: $23^{\circ} 15'$
 SP: $322^{\circ}45''$ PS: $142^{\circ}45''$

- (A) QP: $215^{\circ} 00'$; QR: $87^{\circ} 30'$; RQ: $276^{\circ} 30'$; RS: $201^{\circ} 15'$
 (B) QP: $216^{\circ} 10'$; QR: $93^{\circ} 50'$; RQ: $287^{\circ} 10'$; RS: $208^{\circ} 45'$
 (C) QP: $210^{\circ} 30'$; QR: $76^{\circ} 50'$; RQ: $217^{\circ} 20'$; RS: $210^{\circ} 15'$
 (D) QP: $216^{\circ} 00'$; QR: $97^{\circ} 30'$; RQ: $277^{\circ} 30'$; RS: $203^{\circ} 15'$

Ans : (D)

[PCB Civil 2022]

Q.55 The tangential method of tacheometry uses:

- (A) the readings against the top and bottom cross-hairs only
- (B) the readings against the middle cross-hair only
- (C) a constant intercept on the staff
- (D) the readings against all the three cross-hairs

Ans : (B)

[PCB Civil 2022]

Q.67 Which of the following soil group has excellent workability as construction material?

- (A) SC
- (B) GM
- (C) SM
- (D) GW

Ans : (D)

[PCB Civil 2022]

Q.68 A soil having Specific gravity 'G' equal to 2.66 has a theoretical maximum dry density of 1.865 g/cm³ which corresponds to zero air voids at the OMC. Calculate the corresponding dry unit weight, if the density of water (ρ_w) = 1 g/cm³.

- (A) 18.29 kN/m³
- (B) 21.56 kN/m³
- (C) 22.78 kN/m³
- (D) 24.79 kN/m³

Ans : (A)

[PCB Civil 2022]

Q.69 A granular soil is subjected to major and minor principal stress of 8 MN/m² and 2 MN/m², respectively. What will be the radius of Mohr's circle?

- (A) 6 MN/m²
- (B) 3 MN/m²
- (C) 4 MN/m²
- (D) 10 MN/m²

Ans : (B)

[PCB Civil 2022]

Q.70 Determine the unconfined compressive strength of normally consolidated deposit of clay, if the effective stress at a depth of 15 m is 180 kN/m², plasticity index of the clay is 75% and the undrained cohesion value of clay equal to 69.75 kN/m².

- (A) 34.87 kN/m²
- (B) 139.5 kN/m²
- (C) 69.75 kN/m²
- (D) 208.35 kN/m²

Ans : (B)

[PCB Civil 2022]

Q.71 A dry sand specimen was tested in a triaxial machine with the cell pressure of 50 kPa. If the deviator stress at failure was 100 kPa, calculate the major principal stress at failure.

- (A) 100 kPa
- (B) 150 kPa
- (C) 50 kPa
- (D) 500 kPa

Ans : (B)

[PCB Civil 2022]

Q.72 Laxmi Publications (P) LTD

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Ans : None

[PCB Civil 2022]

Q.73 Which of the following expression is correct to calculate permissible stresses used in working stress method of design?

- (A) Permissible Stress = Yield Stress × Factor of Safety
- (B) Permissible Stress = Yield Stress / Factor of Safety
- (C) Yield Stress = Permissible Stress / Factor of Safety
- (D) Permissible Stress = Yield Stress - Factor of Safety

Ans : (B)

[PCB Civil 2022]

Q.74 Which of the following does NOT belong to the limit state of strength?

- (A) Fracture due to fatigue
(C) Rupture of the structure

- (B) Brittle fracture
(D) Vibrations in structures

Ans : (D)

[PCB Civil 2022]

Q.75 As per IS 800:2007, The thickness of the lacing bar for single lacing should not be less than of its effective length.

- (A) 30
(B) 40
(C) 50

(D) 60 Hints : Civil Engg - Steel Structure -AS_SK_16 Clear_Translation Template_P1El IS 800, 2007,50 Steel Structure Design , Dr. BC Punimia, Laxmi Publication

Ans : None

[PCB Civil 2022]

Q.76 Which of the following is NOT a type of bolted joint?

- (A) Lap joint
(C) Double cover lap joint

- (B) Single cover butt joint
(D) Double cover butt joint

Ans : (C)

[PCB Civil 2022]

Q.77 In lightly stressed structures, where stiffness, rather than strength, controls the design and fatigue or brittle fracture is not a problem, the _____ welds are entirely adequate and generally more economical.

- (A) fillet
(C) plug

- (B) butt
(D) slot

Ans : (A)

[PCB Civil 2022]

Q.78 The reduction in the weld strength as per long joint is done when the length of the welded joint of a splice or end connections in a compression or tension element exceeds _____ times the throat size of the weld.

- (A) 16
(C) 100

- (B) 150
(D) 120

Ans : (B)

[PCB Civil 2022]

Q.79 Which of the following causes fatigue in a structure?

1. Stress due to self-weight
2. Residual stress
3. Large number of loading cycles
4. Wide range of stress variation

- (A) 1 and 2
(C) 2, 3 and 4

- (B) 3 and 4
(D) 1, 2 and 4

Ans : (B)

[PCB Civil 2022]

Q.80 A metal specimen having a diameter of 20 mm is tested in universal testing machine. It was found that the strain in longitudinal, lateral direction was 4.5×10^{-4} and 1.5×10^{-4} respectively. What will be its fraction of Poisson's?

- (A) 0.33
(C) 0.25

- (B) 0.30
(D) 0.5

Ans : (A)

[PCB Civil 2022]

Q.81 Identify the correct statement with respect to the materials used in steel reinforced cement concrete structures

- (A) Cement concrete has poor compressive strength when compare to its tensile strength
(B) Steel is weak in compression
(C) Steel has a good tensile strength
(D) Cement concrete has good tensile strength when compared to steel

Ans : (C)

[PCB Civil 2022]

- (A) As far as possible, the intersection of the taxiway and runway should be avoided.
- (B) Exit taxiways are always placed at right angles to the runways.
- (C) The route for the taxiway selected should provide the shortest practicable distance from the apron to the runway end.
- (D) Taxiways should be arranged so that aircraft taxiing towards the apron do not interfere with the aircraft taxiing for take-off.

Ans : (B)

[PCB Civil 2022]

Q.90 The main function of taxiways is to provide access from the runways to the terminal area and service hangers. Which of the following statements related to the length of a taxiway is/are true or false? 1: Taxiway length depends upon the distance between the apron and entry end or exit end of the runway. 2: To save fuel consumption, the length of the taxiway should be as short as practicable.

- (A) 1 is true but 2 is false
- (B) Both the statements are true
- (C) 1 is false but 2 is true
- (D) Both the statements are false

Ans : (B)

[PCB Civil 2022]

Q.91 The function of exit taxiways is to minimise the runway occupancy by the landing aircraft. The location of exit taxiways does NOT depend on _____.

- (A) Touchdown speed
- (B) Mix of aircrafts
- (C) Size of apron
- (D) Rate of deceleration of aircrafts

Ans : (C)

[PCB Civil 2022]

Q.92 The aircraft are berthed on the aprons before they are loaded and unloaded. Hence, the loading apron is also known as the _____.

- | | |
|-------------------|-------------------|
| (A) landing apron | (B) parking apron |
| (C) service apron | (D) hangar |

Ans : (B)

[PCB Civil 2022]

Q.93 The relatively small aprons placed at a convenient location in the airport for the temporary storage of the aircraft is called _____.

- | | |
|-----------------------|-----------------------|
| (A) the loading apron | (B) the holding apron |
| (C) the holding bay | (D) the parking apron |

Ans : (C)

[PCB Civil 2022]

Q.94 In the case of flexible pavements, ineffective road surface drainage system is mainly due to:

- (A) reduced thickness of wearing coat
- (B) poor quality of subgrade soil
- (C) Poor compaction of pavement layers
- (D) Inadequate cross slope of pavement surface as well as longitudinal slope of the roadside drains

Ans : (D)

[PCB Civil 2022]

Q.95 Holes of 30 to 50 mm diameter are drilled through the CC pavement slab at a spacing of 1.5 to 3m. The sand-cement slurry is grouted under high pressure through these holes until the gaps under the slabs are filled up and the cracked slab starts lifting. This process is called _____.

- | | |
|-----------------|-----------------|
| (A) mud pumping | (B) mud jacking |
| (C) scaling | (D) Ravelling |

Ans : (B)

[PCB Civil 2022]

Q.96 The turning zone is the area of the airport other than the approach area and it is intended for turning operations of the aircraft in case of emergencies like the failure of engine or trouble in the smooth working of aircraft experienced at the start of the take-off. Which of the following statements related to the turning zone of the aircraft is/are true or false?

1. Any object located within a distance of 4.5 km from the airport reference point is considered as an obstruction, if its height exceeds 51 m above the ground or the established airport elevation, whichever is more.
2. An aircraft operates at a higher height in the turning zone, so it is necessary to ascertain the fact that the area of the turning zone should be free from any obstructions.

- (A) 1 is true but 2 is false
 (B) Both the statements are true
 (C) 1 is false but 2 is true
 (D) Both the statements are false

Ans : (A)

[PCB Civil 2022]

Q.97 The runway is usually oriented in the direction of the prevailing winds. The orientation of a runway is selected after a careful analysis of wind velocity and direction and duration over the past years. Match the wind directions used in runway orientation with the definition.

Wind direction Definition

A. Headwind

1. It acts in the perpendicular direction of aircraft

B. Tailwind

2. It acts in the opposite direction of the head of the aircraft

C. Crosswind 3. It acts in the direction of the landing operation

- (A) A- 1, B -2, C - 3
 (B) A-2, B -1, C -3
 (C) A-3, B -2, C -1
 (D) A-2, B -3, C -1

Ans : (D)

[PCB Civil 2022]

Q.98 The ideal orientation of the runway decided from the study of the wind rose diagram may have to be altered or changed because of the following factors.

1. The orientation of the runway demands excessive grading and earthwork.
2. The location of the runway does not create excessive noise nuisance to the surrounding residential areas and public places of importance. Which of the following statements is/are true or false?

- (A) Both the statements are true
 (B) 1 is false but 2 is true
 (C) 1 is true but 2 is false
 (D) Both the statements are false

Ans : (C)

[PCB Civil 2022]

Q.99 Which of the following is an INCORRECT assumption in calculating the basic runway length?

- (A) The airport is at sea level.
 (B) No wind is blowing on the runway.
 (C) The runway has a slope in the longitudinal direction.
 (D) The en-route temperature is standard.

Ans : (C)

[PCB Civil 2022]

Q.100 Given the following aircraft class, mix and average gate occupancy time, the capacity of 10 gates that serve the three classes of aircraft is:

Aircraft class Mix(%) Average occupancy time (min)
 15 20

30 40

55 60

Assume that each gate is available for all aircraft.

(A) 18 aircrafts/hour**(B)** 22 aircrafts /hour**(C)** 12.5 aircrafts /hour**(D)** 5.5 aircrafts/hour**Ans : (C)**

[PCB Civil 2022]

Q.140 Identify the INCORRECT statement with respect to vertical curves used in highways.**(A)** The parabolic curve produces excellent riding qualities and is preferred in highways**(B)** The circular arc produces good riding qualities and is preferred in highways.**(C)** Provision of parabolic curves is not recommended in highways.**(D)** Abrupt change in the rate of grade is not recommended in highways.**Ans : (C)**

[PCB Civil 2022]

Q.141 In setting out a vertical curve, the difference in elevation between a point on the tangent and the corresponding point on the curve varies as:**(A)** the square of its horizontal distance from the point of tangency**(B)** 2 times the horizontal distance from the point of tangency**(C)** 0.5 times the horizontal distance from the point of tangency**(D)** the square root of its horizontal distance from the point of tangency**Ans : (A)**

[PCB Civil 2022]

Q.142 A simple circular curve of radius(r) 150 m is to be set on the ground with a central angle 60° . Calculate its length of long chord.**(A)** 100 m**(B)** 125 m**(C)** 150 m**(D)** 175 m**Ans : (C)**

[PCB Civil 2022]

Q.143 Identify the INCORRECT statement with respect to electromagnetic radiation used in remote sensing.**(A)** It produces a time-varying magnetic field and vice versa.**(B)** Once electromagnetic radiation is generated, it remains self-propagating.**(C)** It is capable of travelling across space.**(D)** Human eyes are sensitive to all the parts of the electromagnetic system.**Ans : (D)**

[PCB Civil 2022]

Q.144 Identify the INCORRECT statement with respect to airborne remote sensing system.**(A)** Very high spatial resolution images (20 cm or less) can be obtained through airborne remote sensing.**(B)** Airborne remote sensing is better than satellite remote sensing to map a large area at a less cost.**(C)** In airborne remote sensing, downward or sideward looking sensors mounted on aircrafts are used to obtain images of the earth's surface.**(D)** The major disadvantage of airborne remote sensing is the high cost per unit area of ground coverage.**Ans : (B)**

[PCB Civil 2022]

Q.145 Which of the following parameters is NOT covered in the design of structures by limit state of serviceability?**(A)** Deformation and deflections**(B)** Vibrations in the structures**(C)** Corrosion and durability**(D)** Equilibrium of the structures**Ans : (D)**

[PCB Civil 2022]

Q.146 As per IS:456-2000, while doing the design for the limit state of collapse in flexure, the maximum strain in concrete at the outermost compression fibre is taken as _____ in bending.

(A) 0.35**(B)** 0.035**(C)** 0.00035**(D)** 0.0035**Ans : (D)**

[PCB Civil 2022]

Q.147 What is the failure pattern of a column with low slenderness ratio, (i.e., relatively short and stocky column) subjected to an axial load which can cause failure?

(A) Fail under ultimate loads, with the material reaching its ultimate strength**(B)** Fail under relatively low compressive loads and susceptible to buckling failure**(C)** Fail due to development of considerable secondary moments**(D)** Fails due to development of both bending and shear stresses**Ans : (A)**

[PCB Civil 2022]

Q.148 As per IS:456-2000, the minimum cross-sectional area of longitudinal reinforcement to be provided in a RCC column of size 200 mm × 600 mm is ____.

(A) 750 mm²**(B)** 960 mm²**(C)** 1060 mm²**(D)** 1250 mm²**Ans : (B)**

[PCB Civil 2022]

Q.149 As per IS:456-2000, in the case of RCC beams, what is the maximum depth of the web, beyond which side face reinforcement shall be provided along the two faces of beam?

(A) 450 mm**(B)** 550 mm**(C)** 650 mm**(D)** 750 mm**Ans : (D)**

[PCB Civil 2022]

Q.150 According to IS:800-2007, the maximum effective slenderness ratio for tension members in which reversal of direct stresses occurs due to loads other than wind or seismic forces is ____.

(A) 100**(B)** 140**(C)** 180**(D)** 220**Ans : (C)**

[PCB Civil 2022]